



Ismail Aissaoui

Dear Admissions Committee,

I am excited to apply for the PhD position in Pedestrian and Cyclist Path Prediction at the University of Southampton. The opportunity to contribute to establishing new models of pedestrian, cyclist, and scooter rider path prediction for complex urban environments resonates deeply with my research interests.

As an AI Data Science Engineer, I have gained experience in developing innovative solutions that leverage machine learning systems. My most relevant achievement is *MedicalMambaTS, a linear-time state space model that enables sub-millisecond bedside inference. T-latency model that can handle real-time data streams.*

The Social Forces approach, which I have studied, is an excellent example of how established models may fail to account for the momentum of objects like cyclists and scooters. My MambaTS model demonstrates the potential to integrate these physical constraints into path prediction models, enabling more accurate forecasts in complex urban environments.

I am confident that my research background and skills will enable me to make significant contributions to this project. I look forward to discussing my application with you further.

Sincerely,